

The Central Distribution: A Minimax-KL Approach to Noninformative Priors and Feature Selection

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1 Introduction

We begin by revisiting a fundamental decision-making problem and its treatment within Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

We assume that some evidence e about the unknown environment state is available, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A | E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg \min_S \sum_a \sum_{x,e} \ell(a, x) S(a | e) P^*(x, e),$$

where P^* is the joint probability distribution of the environment state and the evidence.

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg \min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

A strategy that, for each evidence e , selects the action minimizing the expected loss conditioned on e , is optimal:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg \min_a \mathbb{E}_{P^*} [\ell(a, x) \mid e] \\ &= \arg \min_a \sum_x \ell(a, x) P^*(x \mid e). \end{aligned} \tag{1}$$

Now, suppose that we decide to use a different evidence E' instead of E . Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that, in reality, we cannot freely choose any random variable as our evidence. We can only choose evidence that is a deterministic function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Using a strategy based on a simpler form of evidence resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E' .

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory.

The key point here is that the space of strategies using E' is a subset of the strategies that use E . When we are searching for the best strategy based on E , we are implicitly evaluating all simpler forms as well. This may explain why the standard Bayesian framework does not favor the simpler evidence E' .

Plausibly, if we choose an appropriate prior distribution, when we are finding the optimal strategy, the relevant information should be automatically extracted from the evidence by the Bayesian framework. In other words, an appropriate prior should enable feature selection. However, when applied in a conventional Bayesian setting, many commonly used noninformative priors, such as the flat prior, do not naturally promote feature selection.

In this paper, we introduce the central distribution, a minimax-KL representative of a set (or hierarchy) of probability distributions. We establish its connection to reference priors [1], develop asymptotic central priors, and

show how hierarchical central distributions naturally perform feature selection.

2 Central Distribution

Consider a random variable X . In probabilistic modeling, our knowledge about X is represented by a probability distribution, which serves as a basis for making inferences.

However, in many situations, our information may be too limited to justify choosing a single distribution. Instead, we may only be able to specify a set of plausible distributions. Such a set represents some form of *uncertainty* or *partial knowledge*: although it does not identify a single distribution, it constrains the range of plausible probabilistic models.

A well-known example of this idea is the exchangeability assumption used in de Finetti’s theorem. A sequence of random variables $X_1, X_2, \dots$ is said to be exchangeable, if the ordering of the variables carries no information. That is, the joint probability distribution remains invariant under permutations of the variables. This assumption imposes a constraint, effectively introducing a set of plausible distributions. Thus, exchangeability can be seen as a specific form of partial knowledge.

In this work, we adopt the view that a set of probability distributions can represent a meaningful state of partial knowledge, and that such set-valued uncertainty may itself be used as the basis for inference.

More generally, partial knowledge may possess a hierarchical structure. In the simplest case, uncertainty is represented by a single set of candidate distributions. In richer settings, however, this uncertainty may be organized into nested collections, where elements of a set may themselves be sets of distributions. Such nesting can continue recursively, yielding a hierarchical uncertainty structure.

Crucially, the choice of hierarchical organization is itself informative: grouping distributions into subsets expresses that certain alternatives are regarded as belonging to a common epistemic category and should therefore be aggregated before being compared with alternatives in other groups. Consequently, the hierarchy encodes information beyond the collection of leaf distributions alone.

This hierarchical organization may be interpreted as introducing multiple layers of latent structural decisions, where higher-level choices restrict the admissible lower-level alternatives.

Within a Bayesian framework, inference ultimately requires a single probability measure. Therefore, to use hierarchical partial knowledge in Bayesian

inference, we require a principled method for transforming a nested set-valued uncertainty representation into a single probability distribution while preserving the informational structure encoded by the hierarchy.

We adopt the principle that the probability measure used to represent prior beliefs about the environment should depend only on the available information about the environment itself, and not on agent specific considerations such as the loss function or the informativeness of observations. We refer to this modeling assumption as *the principle of objectivity*.

From a minimax perspective, when uncertainty is represented by a set of candidate distributions, a natural representative distribution is one that minimizes the worst-case divergence to the members of the set. Such a distribution may be viewed as the most conservative or robust single-distribution summary of the available partial knowledge.

Definition 2.1 (Central Distribution). Let X be a random variable and $\mathcal{Q}$ be a set of probability distributions over X . We say a distribution C is a *central distribution* of $\mathcal{Q}$ for X , if for every probability distribution P , we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel C(X)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P(X)], \quad (2)$$

where $\mathbb{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence.

Note that, under this definition, the central distribution of $\mathcal{Q}$ is not necessarily an element of $\mathcal{Q}$. That is, the distribution that best represents the set in the specified sense may lie outside the set it summarizes.

Sometimes partial knowledge is represented by a hierarchical structure of sets of distributions. Our current definition of the central distribution does not account for this case. The following recursive definitions extend the concept to handle such situations. While not directly about feature selection, this extension plays a key role in enabling the connection between central distributions and feature selection, a connection that will become clearer later.

Definition 2.2 (Hierarchy of Distributions). A *hierarchy of distributions* over X is defined recursively as follows:

- Any set of probability distributions over X is a hierarchy of distributions over X .
- Any set whose elements are hierarchies of distributions over X is itself a hierarchy of distributions over X .

Intuitively, a central distribution acts as a representative or summary of a set of distributions. While it may not belong to the set itself, it reflects the overall properties of the set in a way that is meaningful for inference. It also allows probability distributions to be evaluated under partial knowledge, thereby inducing a notion of divergence.

The concept of a central distribution is not intended to yield a unique object; rather, all distributions satisfying the definition can be regarded as equally good representatives. However, when multiple central distributions exist, our constructions should not depend on an arbitrary choice among them. To avoid such dependence, we define the central divergence using the best divergence attained over the set of all central distributions.

Definition 2.3 (Central Divergence). For any distribution P , we define the *central divergence* of P from a hierarchy of distributions $\mathcal{H}$, with respect to a random variable X , as follows:

$$\mathbb{D}_{\text{CD}}[\mathcal{H} \parallel P(X)] = \inf_C \mathbb{D}_{\text{KL}}[C(X) \parallel P(X)],$$

where C denotes a central distribution of $\mathcal{H}$ for X .

So far, this definition applies only when the hierarchy is a single set of distributions. The following definition extends the construction recursively to general hierarchical uncertainty structures.

Definition 2.4. Let X be a random variable, and let $\mathcal{H}^*$ be a hierarchy of distributions over X whose elements are hierarchies of distributions. We say C^* is a central distribution of $\mathcal{H}^*$, if for every probability distribution P we have:

$$\sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_{\text{CD}}[\mathcal{H} \parallel C^*(X)] \leq \sup_{\mathcal{H} \in \mathcal{H}^*} \mathbb{D}_{\text{CD}}[\mathcal{H} \parallel P(X)],$$

where $\mathbb{D}_{\text{CD}}[\mathcal{H} \parallel \cdot]$ refers to the central divergence as defined in Definition 2.3.

A simple example illustrates that hierarchical organization can affect the resulting representative distribution even when the underlying admissible sets remain unchanged.

Let X be a random variable with three possible outcomes x_1, x_2, x_3 . For each i , let S_i denote the set of all probability distributions over X that assign zero probability to x_i .

By symmetry, the central distribution of each set S_i is the uniform distribution over its support:

$$C_{S_1} = \left(0, \frac{1}{2}, \frac{1}{2}\right), \quad C_{S_2} = \left(\frac{1}{2}, 0, \frac{1}{2}\right), \quad C_{S_3} = \left(\frac{1}{2}, \frac{1}{2}, 0\right).$$

If the three sets S_1, S_2, S_3 are treated symmetrically as elements of a single set, then their central distribution is

$$C_{\{S_1, S_2, S_3\}} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

Now consider instead a hierarchical organization in which S_1 and S_2 are grouped together, while S_3 forms its own separate branch. By numerically solving the minimax problem in Definition 2.1, we obtain

$$C_{\{S_1, S_2\}} = (0.25, 0.25, 0.5).$$

Applying the same procedure to the two distributions $C_{\{S_1, S_2\}}$ and C_{S_3} yields

$$C_{\{\{S_1, S_2\}, S_3\}} = (0.4, 0.4, 0.2).$$

Thus, although the underlying admissible sets are identical in both constructions, the resulting representative distribution differs.

This demonstrates that hierarchical structure carries informational content beyond the collection of leaf sets alone. Consequently, flattening a hierarchy into a single unstructured set may discard meaningful epistemic structure.

In many learning problems, uncertainty concerns not only the parameters of a predictive model but also its structural form. In particular, when the relevance of potential features is unknown, each candidate feature subset determines a distinct set of distributions in which the target variable depends only on that feature subset.

Feature selection may therefore be viewed as uncertainty over a collection of distributions. The hierarchical central distribution framework provides a natural mechanism for representing and aggregating such uncertainty. By organizing candidate features hierarchically, one may encode preferences over structural hypotheses, such as favoring sparse subsets, grouping related attributes, or privileging particular interactions.

Candidate feature subsets typically induce overlapping families of admissible distributions. Distributions exhibiting simpler dependency structure may belong simultaneously to many such families, whereas more specific dependency patterns belong to fewer. As a result, hierarchical aggregation through central distributions can induce a preference toward structurally simpler models by assigning more weight to distributions compatible with multiple structural hypotheses.

2.1 Minimax Characterization

With the notion of hierarchical central distributions established, we now develop the theoretical machinery needed to characterize and compute them. The following results provide optimization-theoretic characterizations of central distributions that form the foundation of the subsequent analysis.

The central distribution is essentially the solution to a convex-concave game. Since the solution to a convex-concave game corresponds to a saddle point, by defining the appropriate objective function, we can characterize the central distribution using standard convex optimization techniques.

Definition 2.5 (Saddle Point). Consider a function $f : \mathcal{W} \times \mathcal{Z} \rightarrow \mathbb{R}$. We refer to a pair (w^*, z^*) as a *saddle point* for f , if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Next, we establish the connection between saddle points and central distributions. One significant insight provided by the next theorem is that it suggests inference and searching for a central distribution can be combined and performed simultaneously when finding a saddle point.

Theorem 1. *Let X be a discrete random variable, and $\mathcal{Q}$ be a finite set of distributions over X . Consider the following function:*

$$f(P, R) = \sum_i \mathbb{D}_{\text{KL}}[Q_i(X) \parallel P(X)] R(i),$$

where $Q_i \in \mathcal{Q}$, and P, R are probability distributions.

If (P^, R^*) is a saddle point for f , then P^* is a central distribution of $\mathcal{Q}$ for X .*

Proof. Let C be a central distribution of $\mathcal{Q}$ for X , and (P^*, R^*) be a saddle point of f . Since X is a discrete random variable and $\mathcal{Q}$ is finite, C exists. As C is a probability distribution, the point (C, R^*) lies within the domain of f . By the definition of a saddle point, this implies that

$$f(P^*, R^*) \leq f(C, R^*). \tag{3}$$

Because R^* is a probability distribution, the definition of f implies:

$$f(C, R^*) \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(X) \parallel C(X)] \cdot$$

On the other hand by the definition of saddle point,

$$f(P^*, R^*) = \max_R f(P^*, R) = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P^*(X)].$$

Thus, Equation 3 implies

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P^*(X)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel C(X)].$$

Since C is a central distribution, we must have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P^*(X)] = \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel C(X)],$$

and P^* is also a central distribution of $\mathcal{Q}$ for X . $\square$

Lemma 1. *Let f be the function defined in Theorem 1. The point (P^*, R^*) is a saddle point for f , if and only if there are constants λ, η and a function μ , such that for every i, x the following conditions hold:*

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \quad (4)$$

$$P^*(x) = \sum_i R^*(i) Q_i(x) \quad (5)$$

$$\sum_i R^*(i) = 1, \quad (6)$$

$$R^*(i) \geq 0, \quad (7)$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0. \quad (8)$$

Furthermore, for a fixed probability distribution R , a distribution $\hat{P}$ minimizes $f(\cdot, R)$, if and only if for all x it satisfies

$$\hat{P}(x) = \sum_i Q_i(x) R(i). \quad (9)$$

Proof. Similar to f , we define $\hat{f}$ as follows:

$$\hat{f}(P, R) = \sum_i R(i) \sum_x Q_i(x) \ln \frac{Q_i(x)}{P(x)},$$

where the requirement that P and R be probability distributions is omitted. This enables us to formulate the saddle point of f as the solution to a constrained optimization problem defined for $\hat{f}$.

Let (P^*, R^*) be a simultaneous solution to the following two optimization problems:

$$\begin{aligned}
& \text{minimize} && \hat{f}(P, R) && \text{w.r.t. } P \\
& \text{subject to} && \sum_x P(x) = 1, \\
\\
& \text{maximize} && \hat{f}(P, R) && \text{w.r.t. } R \\
& \text{subject to} && R(i) \geq 0, && \text{for all } i \\
& && \sum_i R(i) = 1
\end{aligned} \tag{10}$$

Clearly, if (P^*, R^*) exists, it will be a saddle point for f .

Both of these optimization problems are convex and differentiable in the sense of convex optimization; see [2]. To establish this, consider that all the constraint functions are affine, and therefore are concave-convex and differentiable.

Additionally, $\hat{f}$ is a differentiable function with respect to both R and P . For any fixed P , $\hat{f}(P, \cdot)$ is an affine function and is concave.

On the other hand, for a fixed solution to the second optimization problem denoted by $\hat{R}$, we can write

$$\hat{f}(P, \hat{R}) = c - \sum_{i,x} \hat{R}(i) Q_i(x) \ln P(x), \tag{11}$$

where c is not a function of P . Since $\hat{R}$ is a solution to the second optimization problem, it satisfies the corresponding constraints and is therefore a valid probability distribution. Hence, for all i, x we have $\hat{R}(i) Q_i(x) \geq 0$, and $\hat{f}(\cdot, \hat{R})$ is the negative of a linear combination of concave $\ln P(x)$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [2]. In other words, If there exists a point (P^*, R^*) that satisfies the KKT conditions for both problems, it will be a saddle point for f , and vice versa.

The KKT conditions are as follows for every x, i and constants λ, η :

$$\sum_i R^*(i) Q_i(x) = \eta P^*(x) \quad (12)$$

$$\sum_x P^*(x) = 1, \quad (13)$$

$$\sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} - \mu(i) = \lambda, \quad (14)$$

$$\sum_i R^*(i) = 1,$$

$$R^*(i) \geq 0,$$

$$\mu(i) \geq 0,$$

$$\mu(i) R^*(i) = 0.$$

By summing Equation 12 over x and using Equation 14, we have

$$\eta = 1.$$

Thus, we get Equation 5 which also implicitly implies Equation 13.

To prove the second part of the lemma, we observe that Equations 12 and 13 are the KKT conditions for the minimization problem in 10, when R is a probability distribution. Therefore, these conditions provide necessary and sufficient conditions for a point to be a minimizer of $f(\cdot, R)$, where R is a fixed distribution. $\square$

Lemma 2. *Let $\mathcal{Q}$ be a finite set of probability distributions over a discrete random variable X . Define the function g as:*

$$g(R) = \sum_i \mathbb{D}_{\text{KL}}[Q_i(X) \parallel R(X)] R(i),$$

where $Q_i \in \mathcal{Q}$, and R is a probability distribution such that

$$R(X, I) = Q_i(X) R(I).$$

Let f be the function defined in Theorem 1. If (P^*, R^*) is a saddle point for f , then the joint distribution $Q_i(X) R^*(I)$ is a maximizer of g .

Proof. Let (P^*, R^*) be a saddle point of f . By Lemma 1, P^* satisfies Equation 5, and thus:

$$g(R^*) = f(P^*, R^*).$$

Now, let R be an arbitrary probability distribution. From the second part of Lemma 1, it follows that $g(R)$ indeed corresponds to the minimum of $f(\cdot, R)$, which implies

$$g(R) \leq f(P^*, R).$$

On the other hand, since (P^*, R^*) is a saddle point, we also have

$$f(P^*, R) \leq f(P^*, R^*).$$

Combining the two inequalities, we get:

$$g(R) \leq g(R^*).$$

Since R was an arbitrary distribution, this shows that $g(R^*)$ is the maximum of g , as desired. $\square$

The following theorem gives a standard sufficient condition for centrality in the KL-minimax setting which is the rationale behind the term “central distribution.” Exactly like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 2 (Centrality). *Let X be a discrete random variable, and let $\mathcal{Q}$ be a finite set of distributions over X . Suppose C is a convex combination of elements of $\mathcal{Q}$, with all weights strictly positive. If for some constant λ and every $Q \in \mathcal{Q}$ we have:*

$$\mathbb{D}_{\text{KL}}[Q(X) \parallel C(X)] = \lambda,$$

then C is a central distribution of $\mathcal{Q}$ for X .

Proof. Lemma 1 provides a set of sufficient conditions for a saddle point. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions. We can replace Equation 7 with the more restrictive $R^*(i) > 0$ condition. By Equation 8, that implies $\mu(i) = 0$, and Lemma 1 conditions can be simplified to

$$\begin{aligned} \sum_x Q_i(x) \ln \frac{Q_i(x)}{P^*(x)} &= \lambda, \\ P^*(x) &= \sum_i R^*(i) Q_i(x), \end{aligned}$$

which must hold for every $Q_i \in \mathcal{Q}$ and some strictly positive probability distribution R^* .

Since C satisfies these condition, Lemma 1 and Theorem 1 imply that C is a central distribution of $\mathcal{Q}$ for X . $\square$

Lemma 3. *Assuming the conditions of Theorem 2 and using the same λ , if P is a convex combination of elements of $\mathcal{Q}$, then there exists some $Q_0 \in \mathcal{Q}$ such that*

$$\mathbb{D}_{\text{KL}}[Q_0(X) \parallel P(X)] \leq \lambda.$$

Proof. We proceed by contradiction. Suppose that for every $Q \in \mathcal{Q}$, we have $\mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)] > \lambda$. This implies that

$$g(P) > \lambda = g(C),$$

where C is the distribution specified in Theorem 2, and g is the function defined in Lemma 2. Note that $g(P)$ is defined, because P is a convex combination of elements of $\mathcal{Q}$.

However, by the conditions of Theorem 2, the distribution C corresponds to a saddle point of f . This leads to a contradiction, since Lemma 2 asserts that a saddle point of f must be a maximizer of g . Therefore, our assumption must be false. $\square$

2.2 Parametric Families and Central Priors

In the previous subsection theorems were proved under the assumption that $\mathcal{Q}$ is finite; however, they can be extended to cases where $\mathcal{Q}$ can be indexed by a finite-dimensional real vector θ :

$$\mathcal{Q} = \{Q_\theta \mid \theta \in \Theta\}.$$

We can treat θ as a random variable and interpret Q_θ as a conditional distribution $Q(x \mid \theta)$. Then, the weight function, $R(i)$, will essentially become a prior distribution over the parameter θ .

Definition 2.6 (Interior Distribution). Let $\mathcal{Q}$ be a set of probability distributions over a random variable X . A distribution R is called an *interior distribution* of $\mathcal{Q}$ if, for every probability distribution P , there exists some $Q \in \mathcal{Q}$ such that

$$\mathbb{D}_{\text{KL}}[R(X) \parallel P(X)] \leq \mathbb{D}_{\text{KL}}[Q(X) \parallel P(X)],$$

Theorem 3. *Any distribution that can be expressed as a convex combination of elements of $\mathcal{Q}$ is an interior distribution of $\mathcal{Q}$.*

Proof. Let $R = \sum_i \alpha_i Q_i$, where $Q_i \in \mathcal{Q}$, $\alpha_i \geq 0$, and $\sum_i \alpha_i = 1$.

Fix an arbitrary probability distribution P . By the convexity of KL divergence in its first argument, we have:

$$\mathbb{D}_{\text{KL}}[R(X) \parallel P(X)] \leq \sum_i \alpha_i \mathbb{D}_{\text{KL}}[Q_i(X) \parallel P(X)].$$

Now, note that the right-hand side is a weighted average of the KL divergences. So, there must be at least one index j such that

$$\mathbb{D}_{\text{KL}} [R(X) \parallel P(X)] \leq \mathbb{D}_{\text{KL}} [Q_j(X) \parallel P(X)].$$

Since $Q_j \in \mathcal{Q}$, the condition in Definition 2.6 is satisfied, and R is indeed an interior distribution of $\mathcal{Q}$. $\square$

Theorem 4. *Let R be an interior distribution of $\mathcal{Q}$. If C is a central distribution of $\mathcal{Q}$, it is also a central distribution of $\mathcal{Q} \cup \{R\}$ and vice versa.*

Proof. This theorem follows directly from the definition of an interior distribution, which implies that for any probability distribution P , we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P(X)] = \sup_{Q \in \mathcal{Q} \cup \{R\}} \mathbb{D}_{\text{KL}} [Q(X) \parallel P(X)]. \quad \square$$

One important consequence of this theorem is that, in order to identify a central distribution within a parametric model, it suffices to restrict attention to the set of likelihood distributions induced by different values of the model parameters.

The next theorem is a standard result that is typically used in the derivation of central distributions.

Theorem 5. *Let X and T be two random variables, and let $\mathcal{Q}$ be a set of distributions over the pair (X, T) . Assume that for every $Q \in \mathcal{Q}$, we have*

$$Q(X, T) = R(X \mid T)Q(T),$$

where R is a fixed conditional distribution. If C is a central distribution of $\mathcal{Q}$ for T , then $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T) .

Proof. Let C be a central distribution of $\mathcal{Q}$ for T . For an arbitrary probability distribution P we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X, T) \parallel R(X \mid T)C(T)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X, T) \parallel R(X \mid T)P(T)]. \quad (15)$$

Since the KL divergence is nonnegative, for every probability distribution P and every $Q \in \mathcal{Q}$ we have

$$\sum_t \mathbb{D}_{\text{KL}} [R(X \mid t) \parallel P(X \mid t)] Q(t) \geq 0.$$

That implies

$$\mathbb{D}_{\text{KL}} [Q(X, T) \parallel P(X, T)] - \mathbb{D}_{\text{KL}} [Q(X, T) \parallel R(X \mid T)P(T)] \geq 0.$$

Taking the supremum over $Q \in \mathcal{Q}$ preserves the inequality, since it holds pointwise for every $Q \in \mathcal{Q}$:

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X, T) \parallel R(X \mid T)P(T)] \leq \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}} [Q(X, T) \parallel P(X, T)].$$

Combining this inequality with Equation 15, we conclude that $R(X \mid T)C(T)$ is a central distribution of $\mathcal{Q}$ for (X, T) . $\square$

The next theorem highlights a formal connection between the central-distribution framework and Bernardo's reference prior methodology [1].

Theorem 6. *Let X be a discrete random variable, and let $\mathcal{Q}$ be a set of probability distributions over X , indexed by a finite-dimensional random vector θ . Let c be a strictly positive prior distribution for θ such that for $Q(x \mid \theta) \in \mathcal{Q}$ we have*

$$c(\theta) \propto \exp \left\{ \sum_x Q(x \mid \theta) \ln c(\theta \mid x) \right\}. \quad (16)$$

Then, $C(x) = \int Q(x \mid \theta)c(\theta)d\theta$ is a central distribution of $\mathcal{Q}$ for X . We also refer to c as a central prior for X .

Proof. From Equation 16, it follows that there exists a constant λ such that

$$c(\theta) = \exp \left\{ -\lambda + \sum_x Q(x \mid \theta) \ln \frac{Q(x \mid \theta)c(\theta)}{C(x)} \right\}.$$

Since $c(\theta)$ is strictly positive, we have

$$\begin{aligned} c(\theta) &= \exp \{ -\lambda + \mathbb{D}_{\text{KL}} [Q_\theta(X) \parallel C(X)] + \ln c(\theta) \} \\ &= c(\theta) \exp \{ -\lambda + \mathbb{D}_{\text{KL}} [Q_\theta(X) \parallel C(X)] \}, \end{aligned}$$

which implies

$$\mathbb{D}_{\text{KL}} [Q_\theta(X) \parallel C(X)] = \lambda.$$

Since this holds for all θ , and c is strictly positive, by Theorem 2 it follows that C is a central distribution of $\mathcal{Q}$ for X . $\square$

The above condition defines an implicit fixed-point characterization of a central prior. This result should be interpreted as a sufficient characterization rather than a general existence or uniqueness theorem.

Although this theorem is proved directly from Theorem 2, Lemma 2 suggests an alternative route to the connection with Bernardo's reference prior methodology. Since reference priors are defined by maximizing missing information, the optimization characterization provided by Lemma 2 also offers another perspective on this relationship.

2.3 Asymptotic Central Distributions

Suppose we are interested in modeling an infinite sequence of random variables. Let S_n denote the first n variables:

$$S_n = (X_1, X_2, \dots, X_n).$$

In many settings, our uncertainty about S_n can be represented by a central distribution. A prominent example arises when the sequence is assumed to be exchangeable. As discussed earlier, exchangeability can be viewed as a form of partial knowledge that defines a set of plausible distributions, from which a central distribution for S_n can be constructed.

However, this central distribution generally depends on the sequence length n . According to the principle of objectivity, our beliefs should depend only on the available information and not on arbitrary aspects of the experimental setup, such as the number of variables under consideration.

A distribution defined over a longer sequence naturally induces marginal distributions over all shorter initial segments. This suggests that, rather than constructing a separate central distribution for each S_n , we should seek a single probability measure over the infinite sequence, from which the distributions of all finite initial segments can be obtained as marginals.

This motivates the construction of a central distribution that is well defined over S_∞ .

Definition 2.7. Let $\{S_n\}_{n=1}^\infty$ be a sequence of random variables and $\mathcal{Q}$ be a set of probability measures, where each element of $\mathcal{Q}$ defines a distribution over each S_n . Assume C_n is a central distribution for S_n . Let C be a probability measure that defines a distribution over every S_n . C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^\infty$, if we have

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}} [C_n(S_n) \parallel C(S_n)] = 0.$$

Intuitively, this definition states that as $n \rightarrow \infty$ the distribution that C induces over S_n becomes increasingly similar to a central distribution for S_n . In other words, C is a central distribution for S_∞ .

Our next goal is to extend Theorem 2 so that it also applies in this asymptotic setting, thereby providing a characterization of central distributions for infinite sequences.

Theorem 7. Let $\{S_n\}_{n=1}^\infty$ be a sequence of random variables. Suppose C is a weighted average of elements of $\mathcal{Q}$ with strictly positive weights, and there exists a sequence of constants $\{\lambda_n\}_{n=1}^\infty$ such that

$$\lim_{n \rightarrow \infty} \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}} [Q(S_n) \parallel C(S_n)] - \lambda_n| = 0,$$

then C is a central distribution of $\mathcal{Q}$ for $\{S_n\}_{n=1}^\infty$.

Proof. For the purpose of this proof, we additionally assume that for each S_n , there exists a central distribution C_n that satisfies the conditions of Theorem 2. That is, there exists a constant μ_n such that for every $Q \in \mathcal{Q}$

$$\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C_n(S_n)] = \mu_n.$$

For each n , by the definition of central distribution, we have

$$\sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] \geq \mu_n, \quad (17)$$

and, by Lemma 3, it also follows that

$$\inf_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] \leq \mu_n. \quad (18)$$

On the other hand, for each n we have

$$\left| \sup_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n \right| \leq \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n|,$$

and

$$\left| \inf_{Q \in \mathcal{Q}} \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n \right| \leq \sup_{Q \in \mathcal{Q}} |\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n|.$$

This implies that the left hand sides of Equations 17 and 18 both converge to λ_n . Now, by the squeeze theorem, the sequence $\{\mu_n - \lambda_n\}_{n=1}^\infty$ must converge to 0. Therefore, for any arbitrary $Q \in \mathcal{Q}$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)} &= \lim_{n \rightarrow \infty} (\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \mathbb{D}_{\text{KL}}[Q(S_n) \parallel C_n(S_n)]) \\ &= \lim_{n \rightarrow \infty} (\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \mu_n + \lambda_n - \lambda_n) \\ &= \lim_{n \rightarrow \infty} (\mathbb{D}_{\text{KL}}[Q(S_n) \parallel C(S_n)] - \lambda_n) - \lim_{n \rightarrow \infty} (\mu_n - \lambda_n) \\ &= 0. \end{aligned}$$

Since C_n satisfies the conditions of Theorem 2, $\mathbb{D}_{\text{KL}}[C_n(S_n) \parallel C(S_n)]$ is a weighted average of $\sum_{s_n} Q(s_n) \ln \frac{C_n(s_n)}{C(s_n)}$, and we can conclude

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}}[C_n(S_n) \parallel C(S_n)] = 0.$$

Therefore, C is a central distribution of $\mathcal{Q}$ for the sequence $\{S_n\}_{n=1}^\infty$. $\square$

We now present an asymptotic version of Theorem 5, applicable to infinite sequences of random variables.

Theorem 8. *Let $\{S_n\}_{n=1}^\infty$ and $\{T_n\}_{n=1}^\infty$ be two sequences of random variables. Let $\mathcal{Q}$ be a set of probability measures, where each $Q \in \mathcal{Q}$ defines a joint distribution over (S_n, T_n) . Assume there exists a natural number M and a fixed conditional distribution R such that, for all $Q \in \mathcal{Q}$ and $n \geq M$,*

$$Q(S_n, T_n) = R(S_n | T_n)Q(T_n).$$

Let C be a central distribution of $\mathcal{Q}$ for $\{T_n\}_{n=1}^\infty$ (Definition 2.7). Define a probability measure C^ by*

$$C^*(S_n, T_n) = R(S_n | T_n)C(T_n), \quad n \geq M.$$

Then C^ is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$.*

Proof. By Theorem 5, for each $n \geq M$ we have

$$C_n(S_n, T_n) = R(S_n | T_n)C_n(T_n),$$

where $C_n(T_n)$ is a central distribution for T_n , and $C_n(S_n, T_n)$ is a central distribution for (S_n, T_n) . Therefore,

$$\begin{aligned} \mathbb{D}_{\text{KL}} [C_n(S_n, T_n) \| C^*(S_n, T_n)] &= \mathbb{D}_{\text{KL}} [R(S_n | T_n)C_n(T_n) \| R(S_n | T_n)C(T_n)] \\ &= \mathbb{D}_{\text{KL}} [C_n(T_n) \| C(T_n)]. \end{aligned}$$

Since C is a central distribution for $\{T_n\}_{n=1}^\infty$ we have

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}} [C_n(T_n) \| C(T_n)] = 0.$$

It follows that

$$\lim_{n \rightarrow \infty} \mathbb{D}_{\text{KL}} [C_n(S_n, T_n) \| C^*(S_n, T_n)] = 0,$$

so C^* is a central distribution for $\{(S_n, T_n)\}_{n=1}^\infty$. □

Assume $\mathcal{Q}$ is a set of distributions that can be indexed by a parameter vector θ , and $\{T_n\}_{n=1}^\infty$ is a consistent estimator sequence for θ . Let $P(\theta | T_n)$ denote the posterior distribution obtained from an arbitrary smooth positive prior.

Under regular parametric conditions, posterior distributions based on consistent estimators become asymptotically insensitive to the choice of smooth positive prior.

Motivated by this phenomenon, we heuristically derive the asymptotic form of a central prior by considering the posterior density evaluated at the point $t = \theta$. A similar asymptotic construction also appears in [1].

Specifically, we define

$$c(\theta) \propto \lim_{n \rightarrow \infty} \frac{P(\theta \mid T_n = t)|_{t=\theta}}{\mu_n}, \quad (19)$$

where $\{\mu_n\}_{n=1}^\infty$ is a sequence of positive constants chosen so that the limit exists for every value of θ .

Let k denote the normalizing constant converting proportionality into equality. Since the logarithm is a continuous function, Equation 19 implies

$$\lim_{n \rightarrow \infty} \left(\ln \frac{P(\theta \mid T_n = t)|_{t=\theta}}{c(\theta)} - \ln \mu_n + \ln k \right) = 0,$$

Since T_n is a consistent estimator, the sampling distribution $Q(T_n \mid \theta)$ becomes increasingly concentrated around θ as $n \rightarrow \infty$. Therefore, we may write

$$\lim_{n \rightarrow \infty} \left(\sum_{t_n} Q(t_n \mid \theta) \ln \frac{P(\theta \mid t_n)}{c(\theta)} - \ln \mu_n + \ln k \right) = 0.$$

Furthermore, under the usual regularity conditions, the posterior distribution becomes asymptotically insensitive to the choice of smooth positive prior. Motivated by this asymptotic prior-insensitivity, we heuristically replace the original prior with $c(\theta)$, we obtain

$$\lim_{n \rightarrow \infty} \left(\sum_{t_n} Q(t_n \mid \theta) \ln \frac{Q(t_n \mid \theta)}{C(t_n)} - \ln \mu_n + \ln k \right) = 0.$$

Letting

$$\lambda_n = \ln \mu_n - \ln k,$$

we observe that, for each fixed θ :

$$\lim_{n \rightarrow \infty} (\mathbb{D}_{\text{KL}} [Q(T_n \mid \theta) \parallel C(T_n)] - \lambda_n) = 0.$$

Under additional regularity assumptions ensuring that this convergence is uniform over θ , the assumptions of Theorem 7 are satisfied. Thus, the above asymptotic argument suggests that $c(\theta)$ acts as a central prior of $\mathcal{Q}$ for the estimator sequence $\{T_n\}_{n=1}^\infty$.

A central distribution over an estimator cannot, by itself, be used to perform inference at the level of observed outcomes. To do so, we must convert

it into a distribution over the primary variables of interest. Theorems 5 and 8 provide a simple method for this conversion.

When dealing with layered ignorance and seeking the central distribution of a hierarchy of distributions, as defined in Definition 2.2, we typically encounter situations in which a given level of the hierarchy contains only finitely many alternatives. In such cases, and under appropriate regularity conditions, if there exists a consistent estimator for the index identifying these alternatives, then the central distribution reduces to an equally weighted average of the corresponding representative central distributions.

Example 2.1 (Binomial Experiment). Consider a binomial experiment consisting of n independent Bernoulli Trials. Let $\mathcal{Q}$ denote the family of probability distributions corresponding to different values of the success probability $\theta \in [0, 1]$, with θ serving as the indexing parameter for $\mathcal{Q}$. Let T_n be the sample proportion of successes in the n trials. Since T_n is a consistent estimator of θ , we can apply Equation 19 to determine a central distribution for the sequence $\{T_n\}_{n=1}^{\infty}$.

Assuming a uniform prior on θ , Bayes' rule gives the posterior distribution

$$P(\theta | T_n = \frac{k}{n}) = (n+1) \binom{n}{k} \theta^k (1-\theta)^{n-k},$$

which is a Beta($k+1, n-k+1$) distribution. Using Stirling's formula, we derive its asymptotic form:

$$P(\theta | T_n = \frac{k}{n}) \sim \frac{(n+1)n^{(n+\frac{1}{2})}}{\sqrt{2\pi}(n-k)^{(n-k+\frac{1}{2})}k^{(k+\frac{1}{2})}} \theta^k (1-\theta)^{n-k}.$$

We now let $T_n \rightarrow \theta$:

$$P(\theta | T_n = t)|_{t=\theta} \sim \frac{n+1}{\sqrt{2\pi n}} \theta^{-\frac{1}{2}} (1-\theta)^{-\frac{1}{2}}.$$

Therefore, if we define

$$\mu_n = \frac{n+1}{\sqrt{2\pi n}},$$

the following limit exists for $\theta \in (0, 1)$:

$$\lim_{n \rightarrow \infty} \frac{P(\theta | T_n = t)|_{t=\theta}}{\mu_n} = \theta^{-\frac{1}{2}} (1-\theta)^{-\frac{1}{2}}.$$

This implies that $c(\theta) \propto \theta^{-\frac{1}{2}} (1-\theta)^{-\frac{1}{2}}$ is a central prior for $\{T_n\}_{n=1}^{\infty}$.

Example 2.2 (Fused Multinomial Model). Let $\{S_n\}_{n=1}^\infty$ be a sequence where each $S_n = (X_1, X_2, \dots, X_n)$ represents n independent trials of an experiment with k possible outcomes, and $\mathcal{Q}$ be a set of probability distributions over S_n indexed by a real vector θ . Let $f : \mathcal{X} \rightarrow \mathcal{Y}$ be a deterministic function that maps each outcome of X_i to a label $Y_i = f(X_i)$, where $\mathcal{Y} = \{1, \dots, m\}$. Assume that for all i and $Q_\theta \in \mathcal{Q}$, if $f(x_i) = f(x'_i)$ we have

$$Q_\theta(x_1, x_2, \dots, x_n) = Q_\theta(x'_1, x'_2, \dots, x'_n).$$

That is, the conditional distribution $Q_\theta(X_i \mid Y_i = y)$ is uniform over the preimage $f^{-1}(y)$.

We can consider $\theta = (\theta_1, \dots, \theta_m)$ to be a probability vector over $\mathcal{Y}$, where each θ_y specifies the probability of observing label y . For a sequence S_n , let n_y denote the number of times label $y \in \mathcal{Y}$ appears. Let $|f^{-1}(y)|$ denote the number of outcomes in $\mathcal{X}$ that are mapped to label y by f . Then the probability of observing S_n under $Q_\theta \in \mathcal{Q}$, denoted $Q(S_n \mid \theta)$, is:

$$Q(S_n \mid \theta) = \prod_{y=1}^m \left(\frac{1}{|f^{-1}(y)|} \cdot \theta_y \right)^{n_y}.$$

Now, consider the estimator sequence $T_n = \frac{1}{n} (n_1, n_2, \dots, n_m)$, which is a consistent estimator for θ . The conditional distribution $Q(S_n \mid T_n)$ is then given by:

$$Q(S_n \mid T_n) = \frac{1}{n!} \prod_{y=1}^m \frac{n_y!}{|f^{-1}(y)|^{n_y}}. \quad (20)$$

Note that this distribution is independent of Q_θ . Therefore, by Theorem 8, a central distribution for S_n can be obtained from a central distribution for T_n .

As $n \rightarrow \infty$, the distribution of T_n converges to a multivariate Gaussian distribution. Using this fact and Equation 19, we can determine a central prior for $\{T_n\}_{n=1}^\infty$:

$$c(\theta) \propto \prod_{y=1}^m \theta_y^{-\frac{1}{2}}.$$

This is a Dirichlet prior and induces a Dirichlet-multinomial distribution for each T_n :

$$C(T_n) = \frac{\Gamma\left(\frac{m}{2}\right) \Gamma(n+1)}{\Gamma\left(n + \frac{m}{2}\right)} \prod_{y=1}^m \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right) \Gamma(n_y + 1)}.$$

Note that C is a central distribution for the sequence $\{T_n\}_{n=1}^\infty$, viewed as a probability measure on the infinite sequence T_∞ . The notation $C(T_n)$ denotes the marginal distribution of T_n .

By Theorem 8 and Equation 20, we can derive a central distribution for $\{S_n\}_{n=1}^{\infty}$, which induces the following distribution over each S_n :

$$C_f(S_n) = \frac{\Gamma\left(\frac{m}{2}\right)}{\Gamma\left(n + \frac{m}{2}\right)} \prod_{y=1}^m \frac{\Gamma\left(n_y + \frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}\right) |f^{-1}(y)|^{n_y}}.$$

3 Illustrative Examples

We now present some examples of central distributions and demonstrate how they can be used for feature selection.

3.1 Attribute Selection

In this example, we consider a classification problem where the goal is to predict a target class, denoted by Y , from a set of observed nominal attributes $(A_1, A_2, \dots, A_n)$. We assume that Y may depend on an unknown subset of the attributes, rather than the full set.

This reflects a form of sparsity in the relationship between the class and the attributes, where irrelevant features are ignored in the true generative model.

We can use the fused multinomial model of Example 2.2 to capture this relationship. As an example, assume that Y depends only on A_2, A_5 . Then, we define the labeling function f as follows:

$$f(A_1, A_2, \dots, A_n, Y) = (A_2, A_5, Y).$$

In a fused multinomial model with this labeling function, Y will be conditionally independent from other attributes, given A_2, A_5 . Additionally, $P(A_1, A_2, \dots, A_n \mid A_2, A_5)$ will be uniform.

For every possible subset of attributes a distinct fused multinomial model of this form is needed. Each model corresponds to a set of plausible distributions. Definition 2.4 provides a way to define a central distribution for situations like this one.

By Theorem 2 we know that this central distribution would be a weighted average of the central distributions of individual fused multinomial models. Since we have a finite number of models, these weights are uniform:

$$C_{Q^*}(S_n) \propto \sum_i C_{f_i}(S_n).$$

To establish this, observe that we can construct a consistent estimator for the model indicator I by checking for the equality of different outcomes. Specifically, the estimator identifies outcomes with equal frequency in the data and returns the indicator for the model that supports this equality.

As shown in Equation 1, the optimal Bayesian decision rule depends on the chosen loss function. In this example, we adopt the zero-one loss, which leads to predicting the label with the highest conditional probability. These probabilities are computed using the central distribution we previously derived.

To see how this approach works in practice, we carried out a synthetic experiment with the following setup. We considered binary attributes and generated each attribute independently with equal probability for 0 and 1. The class label was defined as a deterministic function of a fixed subset of 3 out of 12 total attributes, as shown in Table 1. The same subset of relevant attributes and labeling function were used across all trials.

Table 1: Mappings of 3-bit inputs under functions f_1 and f_2

Input (bits)	$f_1(\text{Input})$	$f_2(\text{Input})$
000	a	A
001	b	B
010	c	A
011	d	C
100	e	D
101	f	D
110	g	D
111	h	E

For each experiment, we generated a training set of size 20 and a test set of size 60. A larger test set was used to reduce variability in performance estimates, taking advantage of the synthetic nature of the data. To assess robustness to irrelevant features, we progressively removed one attribute at a time from the full set of 12, starting with irrelevant ones and stopping just before removing any of the three relevant attributes.

At each step, we trained and tested two versions of the central distribution: one assuming a deterministic label function (CD-Det) and one allowing for a conditional (non-deterministic) label distribution (CD-Prob). Both versions are constructed by aggregating over 2^{12} fused multinomial models, corresponding to all subsets of the 12 attributes. As a baseline, we also trained a decision tree classifier (DTree). All methods were trained on the same training data and evaluated on the same test data. Each configuration was repeated for 100 independent trials, and we reported the average accuracy (correct predictions divided by total test instances) over these trials.

The results, shown in Figures 1 and 2, reveal notable differences in performance depending on the underlying class assignment function. For function f_1 , both deterministic and probabilistic methods achieve high accuracy, with the deterministic version of the central distribution slightly outperforming the other. In contrast, function f_2 is more difficult to predict, leading to a clearer separation in performance between the two methods. This difficulty

Figure 1: Accuracy vs. Number of Attributes (Function f_1 , 20 train samples)

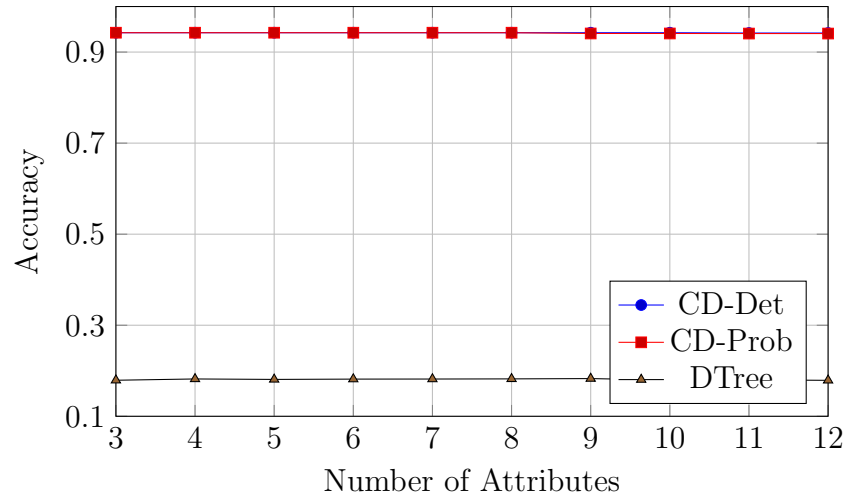
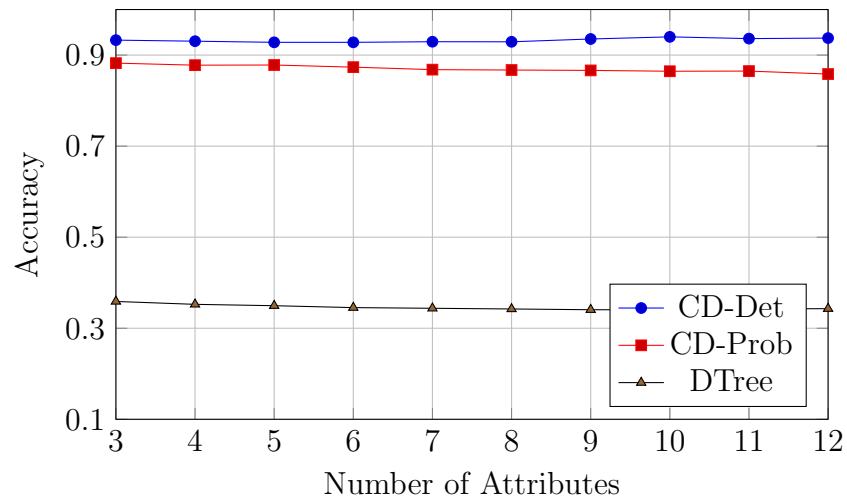
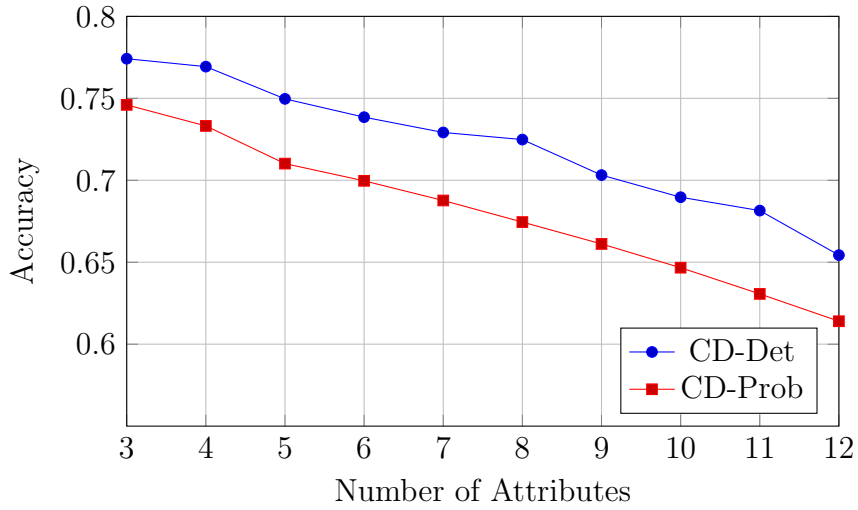


Figure 2: Accuracy vs. Number of Attributes (Function f_2 , 20 train samples)



primarily arises from strong non-deterministic dependencies between subsets of input bits and the assigned label. For instance, when the first input bit of f_2 is 1, the label is often D, creating a misleading probabilistic association. In this case, the deterministic central distribution remains the most accurate overall, while the probabilistic version shows reduced performance due to misleading patterns.

Figure 3: Accuracy vs. Number of Attributes (Function f_2 , 10 train samples)



While removing irrelevant attributes generally led to slight improvements in accuracy, the effect was not substantial in our primary experiments with 20 training samples. However, we observed that the benefit of attribute removal becomes more noticeable when the training set is smaller. In particular, additional experiments with 10 training samples showed a clearer performance gain as irrelevant features were removed, indicating that the impact of irrelevant attributes is more pronounced in low-data regimes (see Figure 3)

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